

Conditional Construct

Problem 1

```
.data
prompt1: .asciiz "Enter first number: "
prompt2: .asciiz "Enter second number: "
msg_gt: .asciiz " is greater than "
newline: .asciiz "\n"

.text
.globl main
main:
    # read first float into $f2
    li $v0, 4
    la $a0, prompt1
    syscall

    li $v0, 6      # read_float
    syscall
    mov.s $f2, $f0

    # read second float into $f4
    li $v0, 4
    la $a0, prompt2
    syscall

    li $v0, 6      # read_float
    syscall
    mov.s $f4, $f0

    # compare: is f2 < f4 ?
    c.lt.s $f2, $f4
    bc1t Print_F4_Greater

Print_F2_Greater:      # f2 >= f4 then f2 is the (>=) larger value
    li $v0, 2
    mov.s $f12, $f2
    syscall

    li $v0, 4
    la $a0, msg_gt
    syscall

    li $v0, 2
    mov.s $f12, $f4
```

```
syscall
j Exit
```

Print_F4_Greater: # f4 > f2

```
li $v0, 2
mov.s $f12, $f4
syscall
```

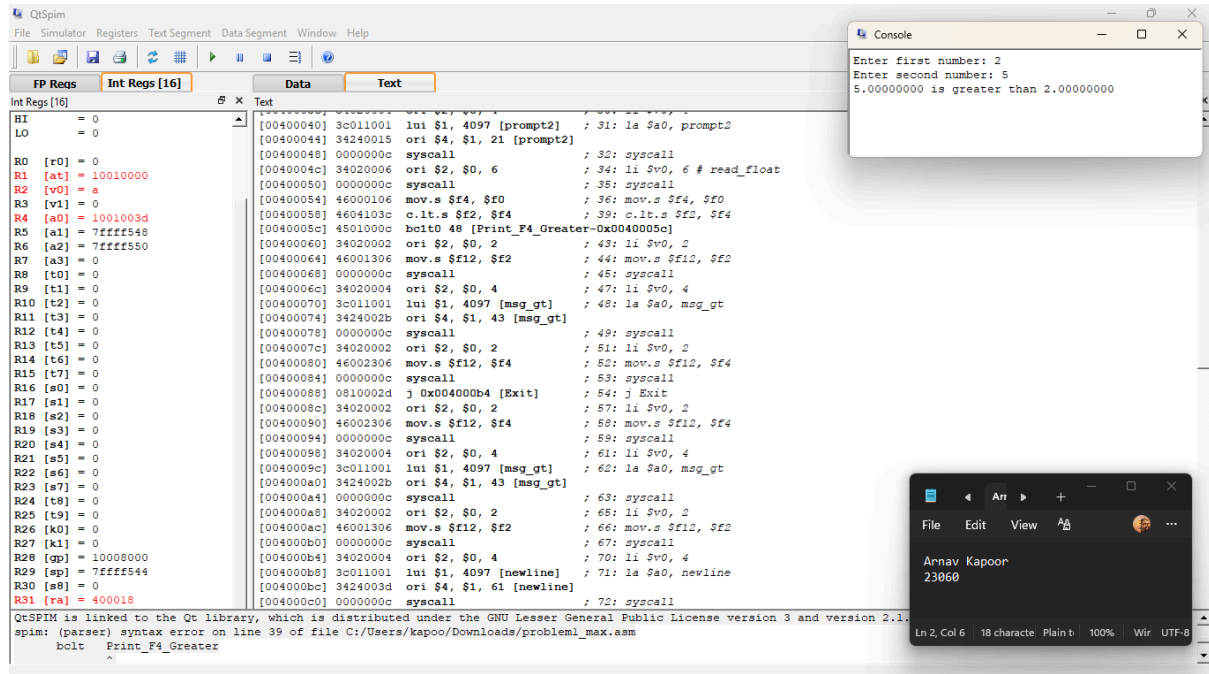
```
li $v0, 4
la $a0, msg_gt
syscall
```

```
li $v0, 2
mov.s $f12, $f2
syscall
```

Exit:

```
li $v0, 4
la $a0, newline
syscall
```

```
li $v0, 10       # exit
syscall
```



Problem 2

```
.data
prompt:    .asciiz "Enter year: "
leap_msg:  .asciiz "Input year is a leap year\n"
notleap_msg: .asciiz "Input year is not a leap year\n"

.text
.globl main
main:
    li $v0, 4
    la $a0, prompt
    syscall

    li $v0, 5      # read_int
    syscall
    move $t0, $v0   # $t0 = year

    li $t1, 4
    div $t0, $t1
    mfhi $t2
    bne $t2, $zero, NotLeap    # year % 4 != 0 -> not a leap year

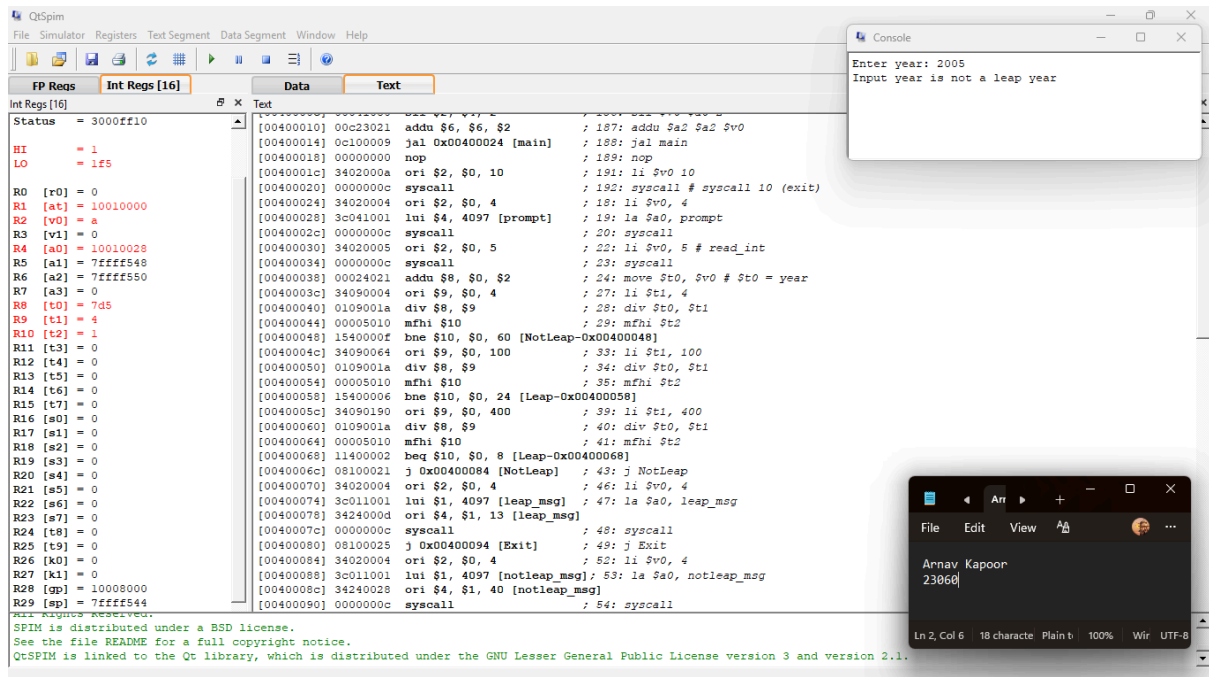
    li $t1, 100
    div $t0, $t1
    mfhi $t2
    bne $t2, $zero, Leap      # divisible by 4 but not by 100 -> leap

    li $t1, 400
    div $t0, $t1
    mfhi $t2
    beq $t2, $zero, Leap      # divisible by 400 -> leap
    j    NotLeap

Leap:
    li $v0, 4
    la $a0, leap_msg
    syscall
    j Exit

NotLeap:
    li $v0, 4
    la $a0, notleap_msg
    syscall

Exit:
    li $v0, 10
    syscall
```



Problem 3

```
.data
prompt_name:    .ascii "Enter student name: "
prompt_held:    .ascii "Enter total number of classes held: "
prompt_attended: .ascii "Enter total classes attended: "
allowed_msg:    .ascii " is allowed to sit in the exam.\n"
notallowed_msg: .ascii " is not allowed to sit in the exam.\n"
name_buffer:    .space 41

.text
.globl main
main:
    # read student name
    li $v0, 4
    la $a0, prompt_name
    syscall

    # read_string
    li $v0, 8
    la $a0, name_buffer
    li $a1, 41
    syscall

    # strip trailing newline from the entered name
    la $t6, name_buffer
StripLoop:
    lb $t7, 0($t6)
    beq $t7, 10, FoundNewline # '\n' == 10
    beq $t7, 0, StripDone
    addi $t6, $t6, 1
    j StripLoop
FoundNewline:
    sb $zero, 0($t6)
StripDone:

    # read total classes held
    li $v0, 4
    la $a0, prompt_held
    syscall
    li $v0, 5      # read_int
    syscall
    move $t0, $v0   # $t0 = held

    # read total classes attended
    li $v0, 4
    la $a0, prompt_attended
    syscall
    li $v0, 5      # read_int
```

```
syscall
move $t1, $v0      # $t1 = attended
```

```
li $t2, 100
mult $t1, $t2
mflo $t3           # $t3 = attended * 100
```

```
li $t4, 75
mult $t0, $t4
mflo $t5           # $t5 = held * 75
```

```
bge $t3, $t5, Allowed
```

NotAllowed:

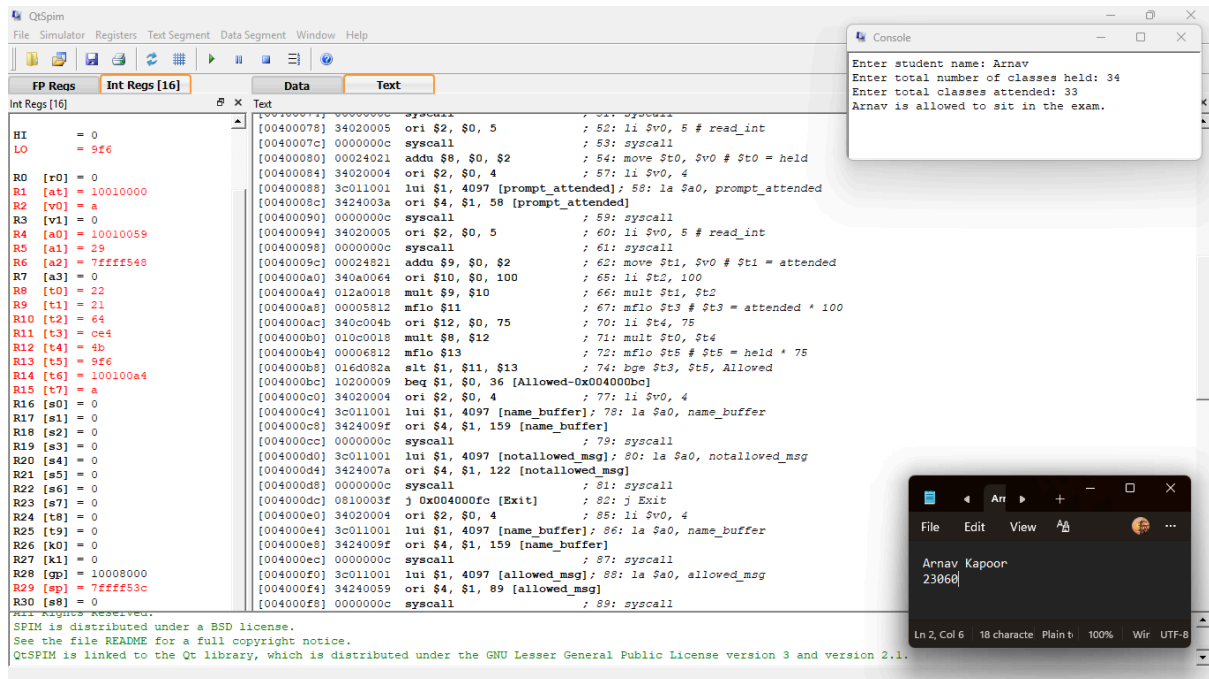
```
li $v0, 4
la $a0, name_buffer
syscall
la $a0, notallowed_msg
syscall
j Exit
```

Allowed:

```
li $v0, 4
la $a0, name_buffer
syscall
la $a0, allowed_msg
syscall
```

Exit:

```
li $v0, 10        # exit
syscall
```



Problem 4

```
.data
prompt: .asciiz "Enter marks (1-100): "
gradeA: .asciiz "Grade: A\n"
gradeB: .asciiz "Grade: B\n"
gradeC: .asciiz "Grade: C\n"
gradeF: .asciiz "Grade: F\n"
```

```
.text
.globl main
main:
    li $v0, 4
    la $a0, prompt
    syscall

    li $v0, 5      # read_int
    syscall
    move $t0, $v0   # $t0 = marks

    li $t1, 80
    bge $t0, $t1, GradeA

    li $t1, 60
    bge $t0, $t1, GradeB

    li $t1, 40
    bge $t0, $t1, GradeC

    j GradeF
```

```
GradeA:
    li $v0, 4
    la $a0, gradeA
    syscall
    j Exit
```

```
GradeB:
    li $v0, 4
    la $a0, gradeB
    syscall
    j Exit
```

```
GradeC:
    li $v0, 4
    la $a0, gradeC
    syscall
    j Exit
```

GradeF:

```
li $v0, 4
la $a0, gradeF
syscall
```

Exit:

```
li $v0, 10      # exit
syscall
```

